



Rainfall retention on the Wye catchment above Belmont

A report prepared on behalf of Wye Salmon Association (WSA)

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Introduction

This report follows up on an earlier report (Green, 2024) in which I examined winter flood trends on the river Wye, using data from the Environment Agency (EA) gauging station at Belmont near Hereford. In this, I suggested that the relatively sudden onset of more frequent floods, between 1980 and 2000, were probably due largely to land management changes occurring over that period – a view apparently shared by the Environment Agency (EA). In this report, I shall look more closely at historical rainfall patterns in the Wye catchment above Belmont, to better understand their impacts on hydrology.

At the global level, climate change is undoubtably causing severe problems. Among other things, we are seeing surface temperatures increasing, ice caps melting, tropical storms becoming more severe, droughts becoming more common and Atlantic circulation currents weakening. But how has this impacted rainfall patterns on the Wye catchment to date? It is basic physics that warmer air can carry more moisture. Consequently, in the UK we are told to expect warmer, drier summers and wetter winters with more severe storm events. Does the available data bear this out so far? More specifically, how has climate change impacted river Wye flooding patterns?

Results & discussion

Rainfall trends

As expected, rainfall data¹ for the catchment above Belmont during the summer months (April to September) exhibits a slowly falling trend. This is shown in figure 1 below. The linear regression gradient of -1.5% per decade equates to a fall of 7.3% over the 50 years period² 1973-2023.

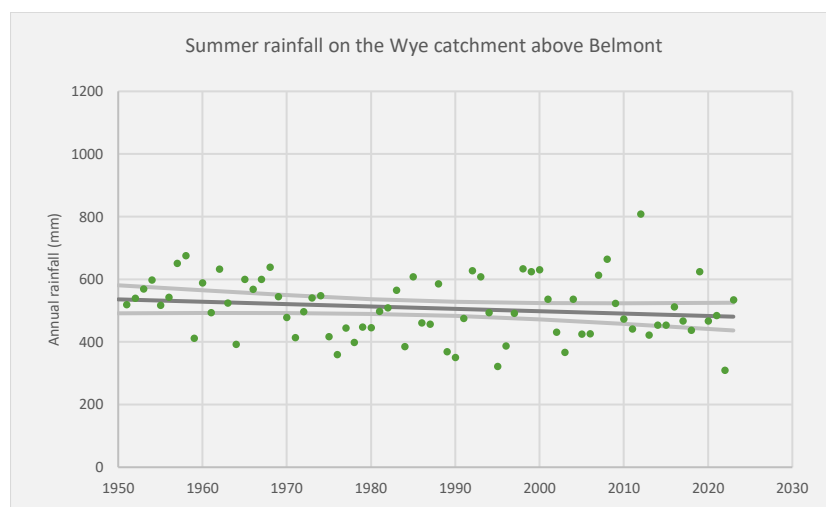


Figure 1: rainfall on the catchment of the river Wye above Belmont during April-September.

The linear regression line (dark grey) has a gradient of -1.5% per decade. Light grey curves show the 95% confidence interval around the regression.

Also as predicted, winter rainfall on the catchment exhibits a rising trend, as shown in figure 2 overleaf. The regression line here has a gradient of 2.7% per decade, or an increase of **14.1% over the period 1973-2023**.

Importantly, both charts show a good fit to a simple, linear regression. There is no suggestion of non-linearity, including in the period 1980-2000 where, in my earlier report (Green, 2024) we found a rapid increase in flood frequency.

¹ Data for 1950-2017 is downloaded from the [CEH - National River Flow Archive](#). Data for 2018-23 has been calculated using [QGIS](#) software applied to gridded *HadUK* rainfall data (Met. Office) downloaded from the [CEDA archive](#) web site.

² This period is chosen because 1973 pre-dates the increase in frequent winter flood frequency and thus provides a useful reference point.

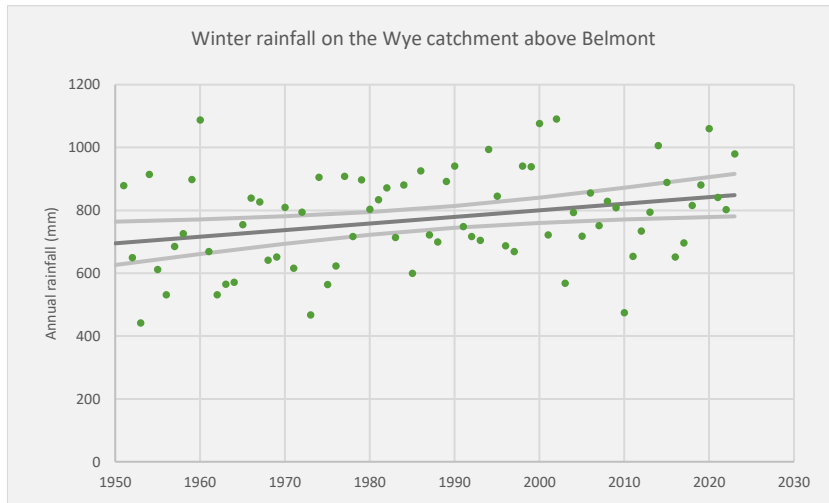


Figure 2: rainfall on the catchment of the river Wye above Belmont during October-March.

The linear regression line (dark grey) has a gradient of 2.7% per decade. Light grey curves show the 95% confidence interval around the regression.

In both of the above charts, we see considerable year to year variance. This simply reflects the variability of the British weather. Moreover, this variability is not increasing; it was always thus. But are individual storms becoming more intense?

Figure 3 shows a histogram of daily rainfall comparing two ten-year periods, 1970-79 and 2014-23. The $\leq 5\text{mm}$ category, clipped in the chart, shows a fall of 107 days in the more recent period. These days have moved into the 10mm to 35mm columns, approximately in proportion to their 1970-79 values.

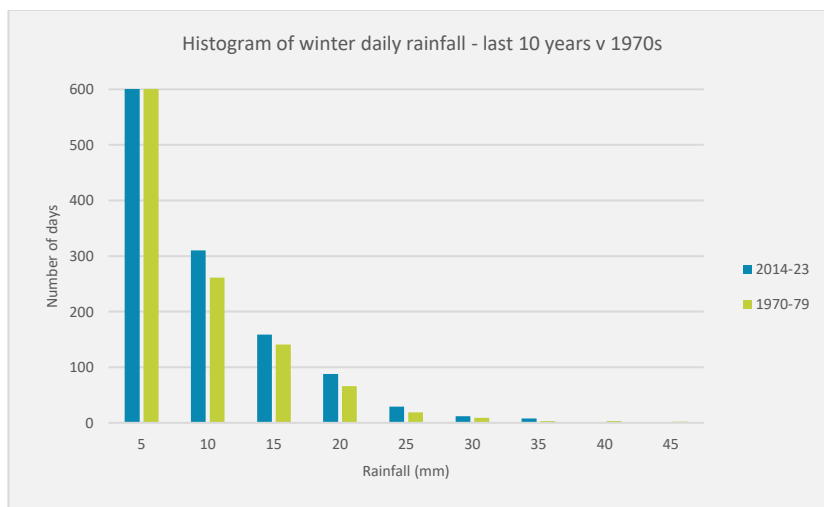


Figure 3: histogram of daily rainfall on the catchment above Belmont, comparing two 10-year periods: 1970-79 & 2014-23.

Note that the 5mm bin is clipped.

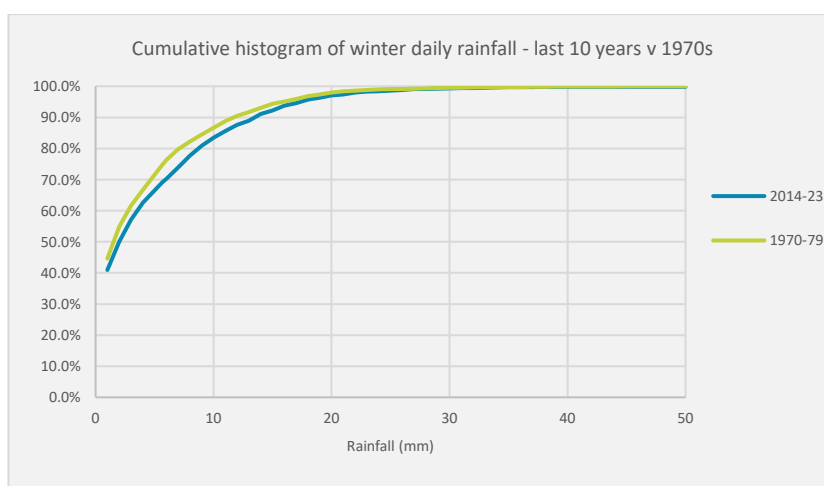


Figure 4: cumulative histogram of daily rainfall on the catchment above Belmont, comparing two 10-year periods: 1970-79 & 2014-23.

We can see this more clearly in the cumulative histogram, figure 4. Fewer dry days leads to a downward shift at the left-hand (low rainfall) end of curve. But the curves are the same shape. There is no evidence of any increase heavy rainfall. The cumulative histogram allows us to extract some helpful statistics, as listed in table 1 below. The 95-percentile values are particularly relevant, in that they show that the number of very wet days (over 15mm could be considered potential flood-causing territory) has increased by considerably less than the 14% increase in total rainfall.

Date range	75%-ile	95%-ile
1970-79	5.7mm	15.9mm
2014-23	7.2mm	17.2mm
Increase	26%	8%

Table 1: summary of flow statistics for two ten-year periods: 1970-79 and 2014-23.

Thus, recent winters have on average 11 fewer dry days than in the 1970s. But these additional wet days approximately follow the same rainfall distribution as the 1970s data. Thus, there appears to be no clear basis for thinking that modern rainfall is any more intense than it was in the 1970s. There are simply 11 more rainy days each winter.

We can further confirm this interpretation by examining the maximum daily rainfall occurring in each year, as in figure 5 below. This chart clearly shows that there is no significant upward trend on peak rainfall since 1950.

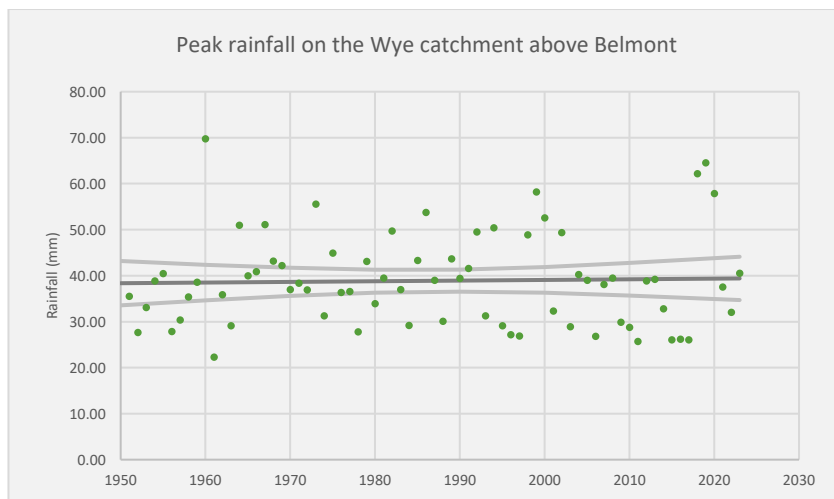


Figure 5: maximum daily rainfall each year on the catchment of the river Wye above Belmont.

The linear regression line (dark grey) shows no significant trend. Light grey curves show the 95% confidence interval around the regression.

Summary:

- There is a large amount of random variance on annual rainfall data which makes it essential to take a long view in any analysis.
- Total winter (October-March) rainfall has increased by 14% over the last fifty years. This rise has been monotonic, with no step changes.
- There is no discernible increase in severity of rainfall over this period. The increased volume of rain appears to be due to more rainy days, not to heavier rainfall per day.

Discharge trends

Figure 6 is reproduced from Green, 2024 and shows the rapid increase in days averaging over 5m at Belmont which occurred during the eighties and nineties. This encapsulates the problem to solve. Five metre floods, meaning the river Wye is overflowing into fields and inundating areas of Hereford, are **five times** more common than they were in the seventies and earlier. Clearly this is not readily explained solely by the rainfall data presented above.

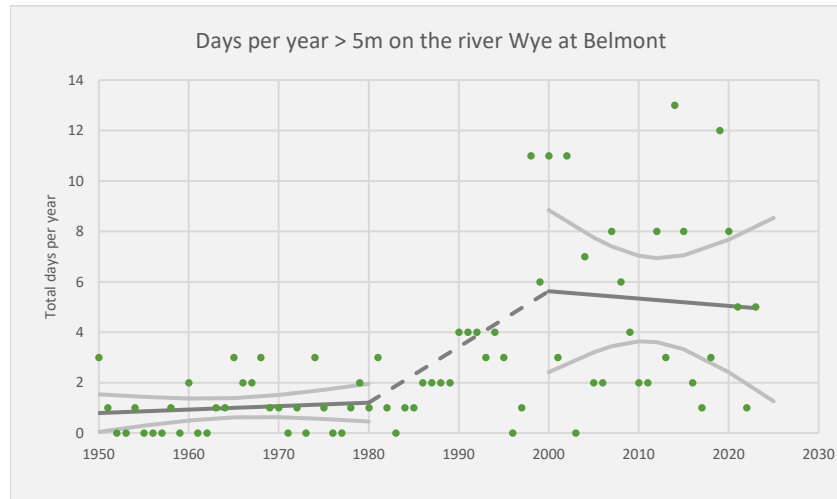


Figure 6: number of days in each year when daily average flow at Belmont exceeded $320 \text{ m}^3/\text{s}$, approximating 5m on the height gauge.

Two regression lines are shown in dark grey, with 95% confidence intervals in light grey.

Let's begin by looking at total winter flow. Figure 7 shows the mean discharge during the winter (October to March) for the river Wye at Belmont³. Since this chart shows a similar step change in the eighties and nineties, I have chosen to fit two separate linear regressions, one for the early period 1950-1980 and one for the recent period 2000-2023. This emphasises the rapid changes occurring in the middle period through the eighties and nineties (dashed in the chart). The result is a massive **36.6% increase in winter discharge between 1973 and 2023**.

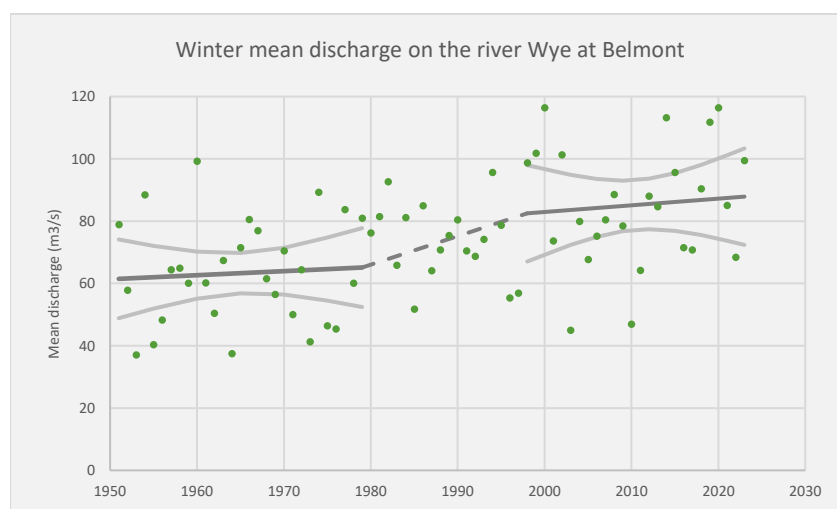


Figure 7: mean discharge on the river Wye at Belmont during the months October-March.

Two regression lines are shown in dark grey, with 95% confidence intervals in light grey.

As presented above for rainfall, it is helpful to view the daily flow data as a cumulative histogram, comparing the same two decades, 1970-79 and 2014-23. This is shown in figure 8 overleaf. The difference between this chart and the equivalent rainfall chart (figure 4) is stark. In figure 8 we see that the winter discharge curve has undergone a large shift to higher flows. We can characterise this shift in terms of flow statistics for the periods 1970-79 and 2014-23 as shown in table 2.

³ This is calculated using daily mean discharge data downloaded from the [DEFRA Hydrology Data Explorer](#).

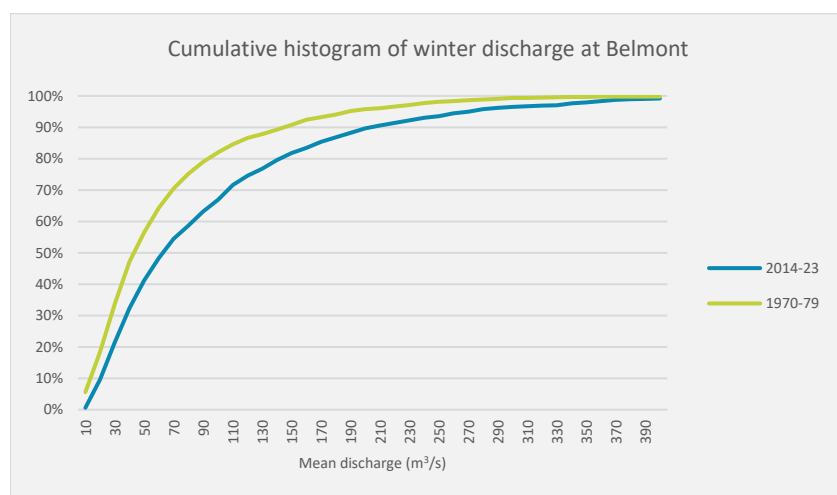


Figure 8: cumulative histogram of mean daily discharge on the river Wye at Belmont during the months October-March, comparing two 10-year periods: 1970-79 and 2014-23.

Date range	Q ₅₀	Q ₀₅
1970-79	43 m ³ /s	190 m ³ /s
2014-23	63 m ³ /s	270 m ³ /s
Increase	48%	42%

Table 2: summary of flow statistics for two ten-year periods: 1970-79 and 2014-23.

There is no surprise that the median value Q₅₀ has increased. However, the increase on Q₀₅ (the 95-percentile discharge exceeded only 5% of the time) is significant. Its value of 190 m³/s in the seventies, approximates a gauge height of about 3.5m – well within the banks. On the other hand, the more recent value of 270 m³/s corresponds to a gauge height of around 4.5m – definitely in flood territory.

Summary:

- The frequency of 5m exceedance at Belmont has increased five-fold since the seventies and earlier. Similar behaviour can be observed for lower thresholds (Green, 2024).
- Total winter (October-March) discharge at Belmont has increased by 37% over the last fifty years. This exceeds the percentage increase in rainfall by a factor of 2.5.
- Both the flood frequency chart and the total winter flow chart exhibit a rapid and significant increase between 1980-2000. This pattern does not match the monotonic increase in rainfall.

Hydrology

The above charts show that the amount of water flowing in the river during the winter months has increased since 1973 by 2.5 times more than the rainfall on the catchment has increased. The fraction of rainfall ending up in the river is known as the hydrologically effective rainfall (HER). Primarily, HER tells us about the catchment characteristics. It is instructive to plot this parameter, as per figure 9 below.

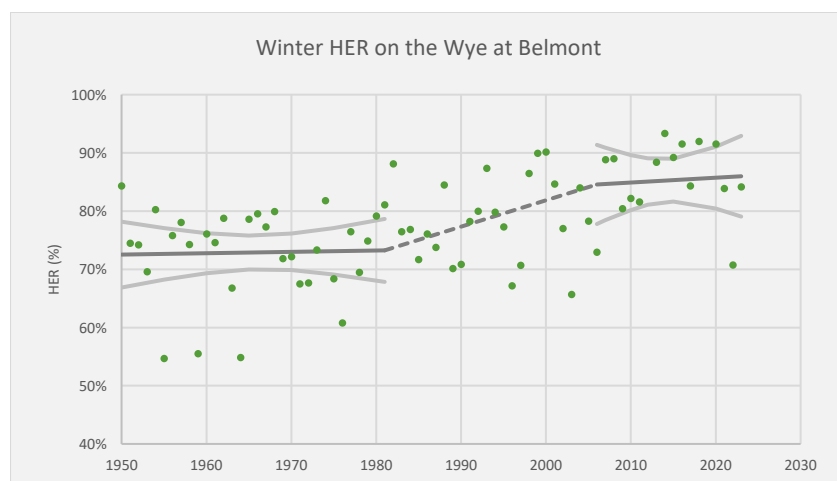


Figure 9: HER on the river Wye at Belmont during the months October-March.

Two regression lines are shown in dark grey, with 95% confidence intervals in light grey.

Through three decades up to 1980, HER remained constant at around 73%. Then it increased rapidly through the eighties and nineties, eventually plateauing at around 85% over the last couple of decades. This represents a huge change in the ability of the catchment to absorb rainfall, which has very little to do with the weather and everything to do with land management practices.

The bar chart in figure 10 concisely summarises the changes which have occurred over the last fifty years. The underlying data is based on the regression analyses presented in figures 2 & 7, not on data for individual years; as such it is statistically robust. The heights of the two bars indicate total rainfall during October-March, which has increased by 14% between 1973 and 2023. Within this, the blue sections indicate the amount of water flowing in the river, which has increased by 37%. The grey sections indicate the amount of water retained within the catchment. It is **half what it was in 1973**.

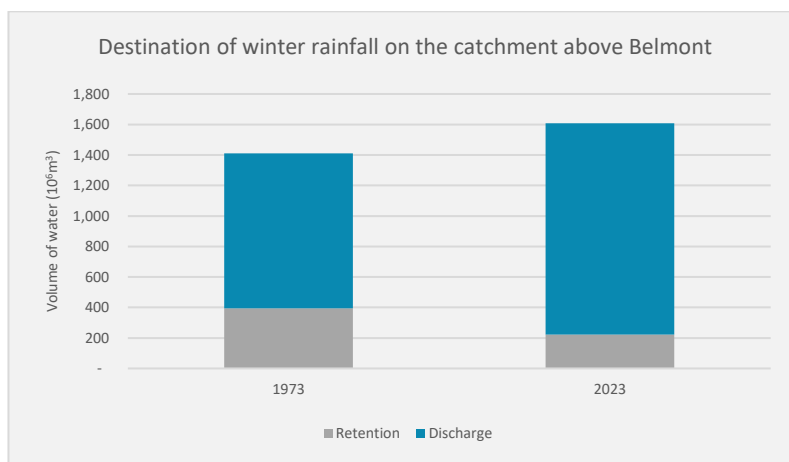


Figure 10: hydrology balance of the river Wye at Belmont.

Bar height represents the measured total volume of rainfall; blue areas represent the measured total volume of water running off via the river. The difference is the grey area, representing the volume of water therefore retained in the catchment soils, or dissipated by evaporation & transpiration.

Thus the 36% increase in river discharge in winter can roughly equally be attributed to increased rainfall (climate change) and a massive reduction in catchment water retention capacity occurring primarily between 1980 and 2000. A crucial point though, is that the two factors play out in different ways. We have seen that the additional rainfall manifests as more wet days, but not as heavier rain. Thus, it will undoubtedly increase the average winter level of the river (by around 14%), but it is not the primary driver of flooding. Increased flood frequency (x5) appears to be driven primarily by the halving of catchment water retention capacity, illustrated by figures 9 & 10. The impact of this change is that amounts of rain which in the 1970s would bring the river up to 3.5m are now bringing it up to 4.5m – thus over-topping the banks.

It is beyond the scope of this report to determine precisely which land management changes are responsible for the reduction in water retention. This is perhaps something to discuss with regulators.

Conclusion

Rainfall on the Belmont catchment has increased over the last fifty years, presumably driven by climate change. However, this increase comes nowhere near explaining the huge rise in winter flood frequency seen over the same period. The rapid onset of flooding through the period 1980-2000 points to land management changes as the primary driver of floods. This hypothesis is strongly supported by the radical change in winter HER which occurred at this time. Catchment water retention in winter appears to be half of what it was in the 1970s and earlier.

References

Green, G. (2024). *Winter flows at Belmont*.